Penny Pals: An Interactive Financial Literacy Application for Children

CSCI 4080U - Design for Educational Applications

Final Project Report

April 12, 2026

Link to Demo: https://edu-app-final-1.onrender.com/dashboard

Abstract

Penny Pals is a web application designed to teach basic financial literacy concepts to children ages 9 to 11. Many traditional learning methods are not engaging enough for younger students, which makes it difficult for them to understand important topics such as saving, spending, and budgeting, which they will be using in the long term. This project aims to address this issue by providing a fun learning experience. This app includes features such as lessons, a coin reward system, an investment game, and a prize shelf to keep users engaged. These features allow the young users to actively make decisions and see the outcomes/consequences, which helps them learn through experience.

Overall, Penny Pals combines learning with gamification to create a more engaging way for children to learn financial literacy.

1. Introduction

Financial literacy is an important life skill, but many children do not learn it at an early age. Most students in elementary school are not taught how to manage money, understand spending habits, or make basic financial decisions. As a result, they often struggle later in life when they are handling real-world financial responsibilities.

To address this problem, our group developed **Penny Pals**, an interactive educational web application that teaches financial literacy concepts to children in a fun, engaging way. The app focuses on helping children understand key ideas such as saving, spending, needs versus wants, and basic budgeting through simple activities and games.

The main goal of Penny Pals is to create an environment where children can practice making financial decisions and see the outcomes or consequences of those decisions. In this app, children will actively interact with the app, earn coins, and use them within the different features of the application. This will help reinforce learning through experience.

2. Background/Related Work

2.1. Educational Context

Financial literacy is becoming more important in today's world, especially as people are expected to make financial decisions at a younger age [1]. However, children often grow up without fully understanding how money works, because they are not taught good financial habits in school.

Children learn best when they are actively engaged. For students between the ages of 9 and 11, learning through interaction, visuals, and simple decision-making activities is alot more effective than passively learning concepts [2]. This is because younger learners have shorter attention

spans and they benefit from immediate feedback. Because of this, it is important for educational tools to be designed in a way that keeps students interested while still teaching meaningful concepts. This is where our interactive application plays a big role.

3. App Design

3.1. Target Users

The target users for Penny Pals are children between the ages of 9 and 11, which typically includes students in grades 4 to 6. At this age, students are starting to develop basic decision-making skills but still prefer learning through visuals and interaction rather than long lectures [3]. These users have shorter attention spans compared to adults, so the app is designed to be simple and easy to navigate. The app design has an emphasis on bright colors, animations, and minimal text so that users are kept interested and focused.

3.2. Learning Objectives

The main goal of Penny Pals is to help children understand basic financial concepts simply and practically. The app focuses on the following learning objectives:

- Identify and classify income and expenses in simulated financial scenarios.
- Distinguish between needs and wants and apply this understanding when making financial decisions.
- Set and monitor short-term savings goals within the application.
- Analyze the outcomes of financial decisions presented in interactive scenarios.
- Reflect on financial choices and adjust decisions based on feedback and repeated attempts.

Instead of just explaining these ideas, the app allows users to practice them through different activities, which helps reinforce learning.

This app also focuses on helping students learn through practice instead of just reading information. Instead of showing long explanations, the app uses short activities and scenarios where users can apply what they learn. This makes it easier for students to understand concepts and remember them over time.

3.3. System Overview

Penny Pals is designed as a web-based application with multiple connected pages that work together to create a full learning experience. The main pages of the app include the Home page, Lessons section, Investment Game, and Prize Shelf. Users can navigate between pages using simple buttons and clear layouts. The design keeps everything consistent, so users do not get

confused when moving between different parts of the app.

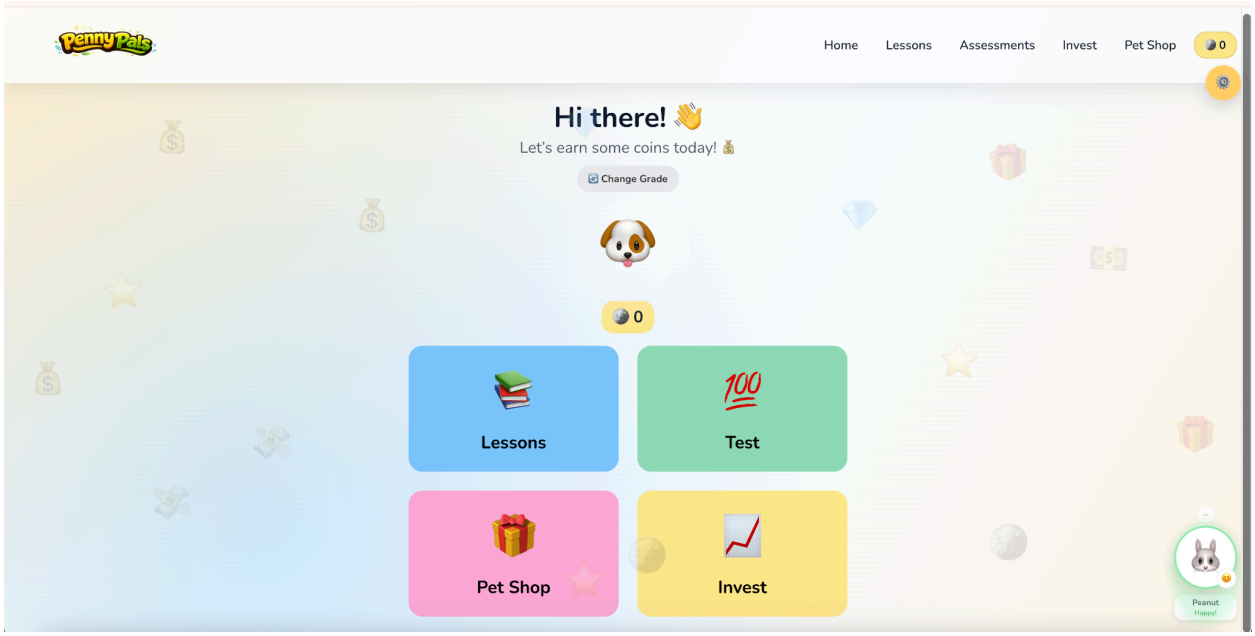

Figure 1: Home page of Penny Pals showing the navigation and main features

3.4. Core Features

Lesson System

The lesson system introduces users to financial concepts in a simple and interactive way. Each lesson focuses on a specific topic, such as needs vs wants or saving money. When users complete a lesson, they are rewarded with coins. For example, a user may earn around 10 coins per lesson.

The lessons are designed to be short and focused instead of long and overwhelming. This follows a microlearning approach, where information is broken into smaller parts so it is easier for users to understand and stay engaged.

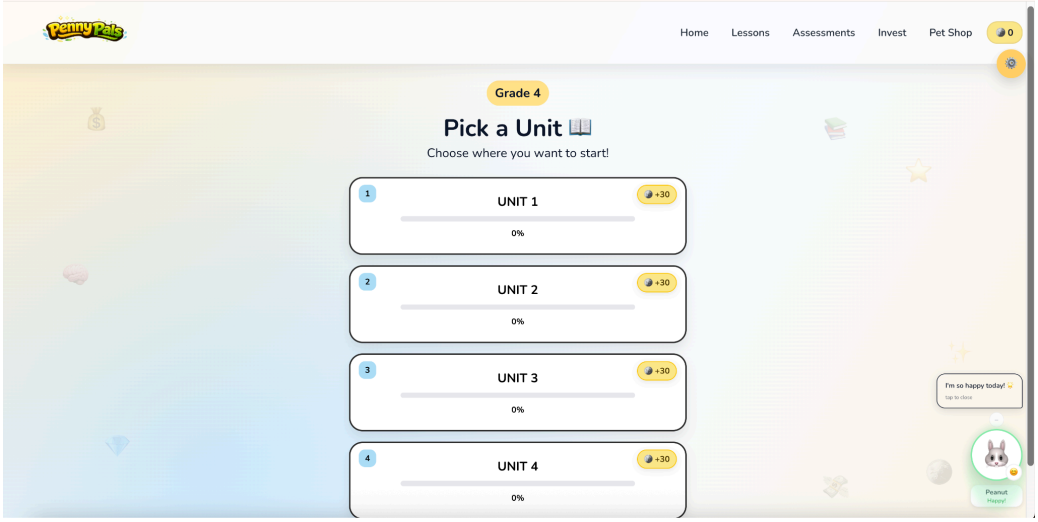

Figure 2: Example of an interactive lesson page within the application

Investment Game

The investment game is one of the main interactive features of the app. In this feature, users can use their coins to invest in different goals and work towards earning different items for their pet.

This system teaches users the idea of making good decisions over time and saving money over time instead of spending it immediately on "fun" things. They learn the importance of keeping a balance between spending money on their "fun meter" and their financial goal.

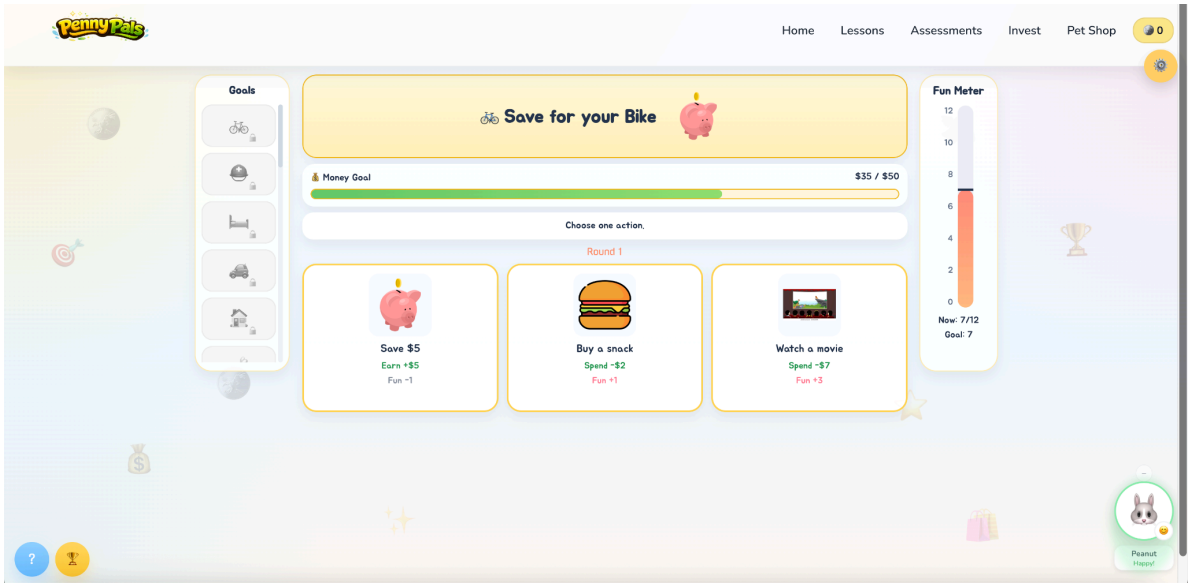

Figure 3: Investment game interface where users allocate coins towards goals

Coin Economy

The app uses a central coin system that connects all features together. Coins are earned through completing lessons and can be used in different parts of the app.

Prize Shelf

The Prize Shelf allows users to view the unlocked rewards/items that they earned through the investment game. These unlocked rewards give the users something to work towards and collect. This feature helps reinforce positive behaviour.

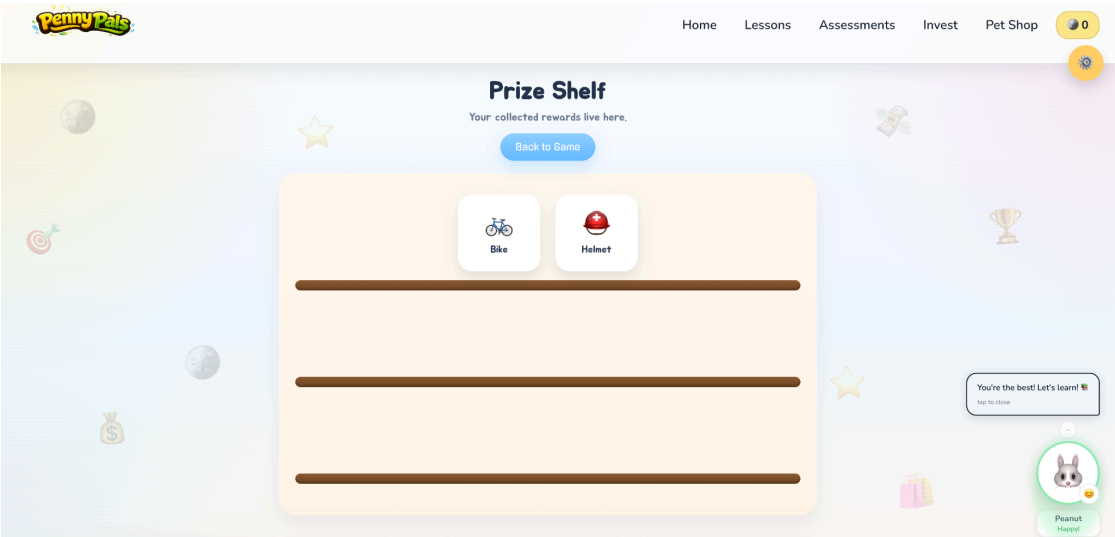

Figure 4: Prize Shelf showing unlocked rewards that users achieved

Gamification Elements

Penny Pals includes several gamification features to make learning more engaging. These include rewards, progress tracking, and immediate feedback. For instance, users can see their progress in the investment game, earn coins after completing lessons, and unlock rewards over time.

A key gamification element is the inclusion of a personalized in-game avatar or pet. Children often form emotional connections with virtual characters, which can support long-term engagement and a sense of responsibility [4]. Allowing learners to customize and care for their avatar using in-game currency earned through learning activities strengthens the connection between effort and reward, while maintaining focus on educational goals.

3.5. UI/UX Design

The user interface of Penny Pals is bright, colorful, and easy to use. Since our target audience is children, the app avoids large blocks of text. We instead focus on images and simple interactions.

Buttons are easy to understand, and the app also uses consistent backgrounds across different pages to create a smooth experience. Moreover, visual effects, such as moving backgrounds, are used to make the app feel more interactive and fun for children.

Another important idea used in the app is learning through mistakes. Instead of penalizing users for wrong answers, the app allows them to try again and provides feedback. This helps users feel more comfortable making decisions and improves their understanding over time.

3.6. Universal Design for Learning (UDL)

To support accessibility and inclusivity, the application follows Universal Design for Learning (UDL) principles. UDL emphasizes providing multiple means of engagement, representation, and expression to accommodate diverse learners [5]. By incorporating visual content, interactive tasks, flexible pacing, and learner choice, the application supports a wide range of learning preferences and abilities.

These design choices ensure that the application remains accessible to learners with varying cognitive strengths, attention levels, and learning needs.

3.7. Design Principles

The design of the app is based on how children learn best. One important principle used is immediate feedback, where users quickly see the results of their actions. This helps reinforce learning and keeps users engaged. Another principle is positive reinforcement, where users are rewarded for completing tasks. This encourages them to continue using the app.

4. Evaluation Plan

4.1. Goals of Evaluation

The main goal of this evaluation is to determine whether Penny Pals is effective in teaching basic financial literacy concepts to children. We also want to see if the app is engaging and easy to use for the target age group. The target participants for this evaluation are children between the ages of 9 and 11. Ideally, the app would be tested with a small group of students within this age range.

4.2. Participants

The target participants for this evaluation are children between the ages of 9 and 11. Ideally, the app would be tested with a small group of students within this age range.

4.3. Methodology

Participants will be asked to use the Penny Pals app for a set period of time, such as 20 to 30 minutes. During this time, they will complete lessons, interact with the investment game, and explore other features like the Prize Shelf. While they are using the app, observations will be made on how they interact with the features. This includes how easily they are able to navigate the app, how long they stay engaged, and how they respond to the rewards and feedback. After using the app, the users will be asked simple questions to understand what they learned.

4.4. Success Metrics

The success of the app will be measured based on 2 key factors. First, users should be showing an improved understanding of the basic financial concepts taught in the app, such as saving money and distinguishing between needs and wants. Secondly, users should stay engaged with the app for the duration of the time they are using it and not lose interest. That will show us how successful the app design is.

5. Conclusion

5.1. Contributions

In this project, we developed Penny Pals, an educational application designed to teach financial literacy to children between the ages of 9 and 11. The app focuses on making learning simple and engaging through visuals, rewards, and interactive features and games.

One of the main contributions of this project is the combination of learning and games. Instead of only presenting information, the app allows the young users to actively make decisions, earn coins, and use those coins in different features such as the investment game. This helps users learn by doing the work rather than just reading. Another key contribution is the design of a child-friendly UI. The app uses bright visuals, simple navigation, and minimal text to make it easier for younger users to understand.

Overall, the project demonstrates how financial literacy concepts can be introduced to younger audiences using interactive technology and interfaces.

5.2. Future Work

There are still several ways the app can be improved in the future. One possible improvement is adding more lessons that cover more financial topics, such as earning money or taxes. The investment game can also be expanded to include more options and scenarios. Another improvement would be adding features that track user progress over a longer period of time, so

users can see how their learning develops over the long term. Additionally, the application can also be developed into a mobile version so that it will be more accessible for all kinds of devices.

References

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